Reconstruction of Visual Decoding via EEG Embeddings with Guided Diffusion

A Test on Reproducibility, Generalizability, and Robustness

Tiago Alexander Reiter
Chair of Data Processing, Technical University of Munich
tiago.reiter@tum.de

***Abstract*—This work investigates the reproducibility, generalizability, and hyperparameter sensitivity of the Adaptive Thinking Mapper (ATM), a neural encoder, specifically tailored at decoding and reconstructing visual stimuli from electroencephalography (EEG) signals with the use of guided diffusion and multimodal embeddings. The ATM framework is based on a two-stage guided diffusion process, which uses a Vision Transformer (ViT) backbone to synthesize images from EEG embedded images. The model is trained and evaluated using their Top-1 and Top-5 accuracy in image retrieval tasks, as well as their performance in k-way classification tasks. The results indicate that although the essential framework proved reproducible under controlled conditions, performance decreased substantially upon transfer to a different dataset. A hyperparameter sweep revealed stable performance within optimal ranges, though decoding accuracy declined notably outside them, indicating the model requires careful tuning. The findings highlight a gap between strong in-domain performance and real-application feasibility and indicate the necessity for future efforts to focus on modular model design, wider dataset benchmarking, and thorough cross-domain testing.**

***Keywords*—Neural Decoding, TUM, ATM Encoder, Research Internship, Chair of Data Processing.**

I. Introduction

Neural decoding, the science and art of translating brain signals to meaningful visual imagery, is no longer the province of theoretical conjecture but a feasible scientific and technological target. Advancements in machine learning, specifically in deep learning, have made recent progress in decoding visual stimuli from brain activity measured using non-invasive means such as functional magnetic resonance imaging or electroencephalography (EEG). EEG is increasingly favored over the alternative in BCIs since it is cheap, yet portable, and enjoys a higher temporal resolution than its competitors. Nevertheless, EEG-based visual decoding faces inherent limitations such as low signal-to-noise ratios (SNR), low spatial resolution, and high variability between and within individuals and recording experiments. Overcoming those challenges is still the focus of research on developing realistic and usable BCI applications. [1] This work investigates the Adaptive Thinking Mapper (ATM), a neural encoder proposed by Li et al. [2], specifically tailored at decoding and reconstructing visual stimuli from EEG signals with the use of guided diffusion and multimodal embeddings. By repeating and supplementing the initial ATM experiments, this work checks three core model evaluation aspects: its reproducibility, generalizability across a different dataset varying in size and difficulty, and sensitivity to changes in training setups and hyperparameters. This work aims to contribute insights towards the practical value and the limitations of current EEG decoding techniques and towards future research and applications in brain-computer interface and neural decoding methods. This work is organized as follows: Section II summarizes related work in EEG-based visual decoding and recent developments in retrieval/classification and generation architectures. Section III shortly explains the proposed model and architecture. Afterwards, Section IV outlines the study's methodology for reproducing the original ATM experiments, evaluating the model's generalizability on an external EEG dataset, assessing its robustness to key hyperparameter changes, and documenting encountered challenges. Section V shows the experimental results comparing performance across conditions. Section VI discusses these results, and Section VII concludes the report.

II. Literature Review

Recent advances in BCIs have sparked growing interest in decoding visual information from neural signals, particularly using EEG. It provides significant practical advantages for decoding applications, including high temporal resolution, portability, and relatively low cost compared to alternatives like fMRI [3]. Nevertheless, EEG signals inherently suffer from low signal-to-noise ratios, inter-subject variability, and susceptibility to artifacts, making preprocessing a critical step. Typical preprocessing techniques include artifact rejection, baseline correction, normalization, bandpass filtering, downsampling, and targeted electrode selection, particularly focusing on occipital and parietal electrodes that encode visual information [4]. Recent research in EEG-based visual decoding has primarily focused on two core tasks: classifying visual stimuli from brain activity and reconstructing visual content based on neural representations.

Modern classification architectures heavily rely on deep neural networks tailored to the structure of EEG data. One prominent example is EEGNet, which uses depthwise and separable convolutions to efficiently extract spatial and temporal features. Although EEGNet was explicitly designed to be compact and computationally efficient—sometimes at the expense of accuracy—it is widely used as a backbone or a primary

encoder in larger EEG decoding architectures [5]. Guenther et al. demonstrated that, when applied to image classification and reconstruction from low-density EEG, the EEGConformer—a hybrid model integrating convolutional modules with transformer layers—achieved a mean test accuracy of 21.2%, whereas EEGNet attained 34.4% on the same task. Similarly, the Adaptive Thinking Mapper (ATM) model proposed by Li et al. integrates spatial-temporal convolutional modules with transformer-based attention mechanisms, achieving top-1 accuracies of approximately 29% for 1,854-class visual stimuli on the THINGS-EEG2 dataset, significantly outperforming previous models like EEGNet and EEGConformer under comparable conditions [2]. The NICE model (Neural Image-Conditioned Embedding) proposed by Song et al. [6] combines EEG embeddings with visual and textual features using a joint similarity learning objective. It achieves top-1 classification accuracies of approximately 19.3% on the THINGS-EEG2 dataset, with Top-5 retrieval reaching around 49.3%. In contrast, ShallowFBCSPNet developed by Lai et al.[7], a traditional architecture based on filter bank common spatial patterns and shallow convolutional layers, performs notably lower, with reported top-1 accuracies of around 6.1% and Top-5 scores near 16.5% on the same dataset.

Reconstructing visual stimuli directly from EEG signals is significantly more challenging than classification. It requires models to categorize visual stimuli and generate precise visual representations from inherently noisy neural signals. Recent advances in generative modeling and multimodal embeddings have greatly enhanced EEG-based reconstruction tasks. Early generative methods, such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), demonstrated initial feasibility but typically yielded blurry or inconsistent reconstructions due to EEG's high noise levels and limited spatial resolution. For instance, Özçelik et al.[8] employed GAN-based architectures for EEG-based image reconstruction but faced challenges in achieving clarity and semantic coherence, resulting in relatively low-quality reconstructions. More recent efforts have adopted multimodal strategies such as CLIP (Contrastive Language-Image Pretraining)[9], leveraging contrastive embeddings to map EEG signals into shared representation spaces with images and/or text. Such embedding-based approaches provide substantial improvements over earlier generative models, achieving sharper reconstructions and notably better semantic coherence. In recent work, latent diffusion models (LDMs) have become central to EEG-based visual reconstruction due to their ability to generate high-fidelity images from noisy brain signals. Guenther et al. [10] proposed a framework that integrates an EEG encoder—adapted from the best-performing EEGNet classifier—with a pretrained LDM. The EEG encoder replaces the classifier head with two 512-unit linear layers, generating embeddings that condition the cross-attention mechanism and the temporal modules of the LDM. Image generation is performed using the Pseudo Linear Multi-Step (PLMS) scheduler over 250 steps. The approach is evaluated using a 50-way top-1 accuracy metric, achieving 35.3% on the best-of-five image generations and 30.9% on the mean accuracy. Alternative reconstruction architectures include EEG2Image, introduced by Singh et al. [11], which uses a generative adversarial network (GAN) trained via min-max optimization. This method is assessed using the Inception Score, achieving 6.78 across ten visual classes. Recently, Lan et al. [12] presented NEUROIMAGEN, a CLIP-LDM-based visual brain signal reconstruction pipeline. Their system embeds EEG signals via CLIP and feeds them into an LDM for generation, reaching a 50-way top-1 accuracy of 85.6% and an Inception Score of 33.5, showing a substantial improvement in both fidelity and semantic alignment. However, despite notable improvements in visual quality and semantic clarity, significant challenges remain in ensuring model robustness and generalization. EEG decoding performance is susceptible to variations in preprocessing pipelines, hyperparameter choices, and architectural configurations [13]. Moreover, performance often drastically decreases when models trained on one dataset are applied to another, more complex or naturalistic visual domain.[4].

III. Experimental Setup

This section provides a concise overview of the ATM model and its architectural components.

A. Model and Architecture

The architecture analyzed in this work is based on the framework proposed by Li et al. [2], titled *Visual Decoding and Reconstruction via EEG Embeddings with Guided Diffusion*. This end-to-end, zero-shot neural decoding framework is designed to classify, retrieve, and reconstruct visual stimuli directly from EEG signals. It consists of three core components: an EEG encoder (the ATM), a contrastive training objective, and a two-stage diffusion-based image generation process.

1. EEG Encoder – Adaptive Thinking Mapper (ATM): ATM-S and ATM-E are specialized implementations of the ATM, a custom EEG encoder built to forward-project brain signals to a common subspace consistent with image embeddings, e.g., the CLIP model embeddings. They are intended to be versatile and swappable modules in the context of EEG-based visual decoding and synthesis. The basic ATM structure relies upon a mix between a Transformer encoder at the channel level, a Temporal-Spatial convolution module, and a multilayer perceptron (MLP) structure. In contrast to other typical techniques, it processes the initial EEG signals directly without the need for segmentation and considers each sequence a patch. The two models differ primarily through their temporal–spatial convolutional blocks since ATM-S uses a ShallowNet–based module while ATM-E utilizes an EEGNetV4–based architecture.

2. Image Embedding – CLIP: The framework uses the CLIP model to generate image embeddings, which serve as the target space for EEG alignment. CLIP embeddings are extracted using a Vision Transformer (ViT) backbone, which has shown superior performance in downstream decoding and generation tasks.

3. Contrastive Training Objective: The training process aligns EEG and image embeddings using a multimodal contrastive learning approach:

$$\mathcal{L} = \lambda \cdot \mathcal{L}_{CLIP} + (1 - \lambda) \cdot \mathcal{L}_{MSE} \tag{1}$$

- $\mathcal{L}_{CLIP}$: Encourages alignment of matched EEG-image pairs while separating non-matching pairs.
- $\mathcal{L}_{MSE}$:s Ensures consistency of the EEG-to-image embedding mapping for generation tasks.
- λ: Balances classification/retrieval and generation objectives.

4. Zero-Shot Inference: Once trained, the ATM encoder supports zero–shot classification, retrieval, and image reconstruction out of the box, with no additional fine-tuning required. This capability arises from its alignment with CLIP's comprehensive semantic embedding space.

5. Two-Stage EEG-to-Image Generation: The model synthesizes the images from EEG embeddings through a two-stage guided diffusion approach. Stage I uses a lightweight diffusion model to predict CLIP image embeddings from EEG embeddings. This prior diffusion model is trained using a U-Net model with classifier-free guidance to find the optimal point between image fidelity and diversity. Stage II sends the predicted CLIP embeddings through a pre-trained SDXL model via an IP-Adapter to guide the image synthesis. The SDXL U-Net embeds the embeddings through cross-attention to produce the final image.

6. Optional Pipelines: The framework supports two optional reconstruction enhancements during Stage II. The Low-Level Pipeline aligns EEG features with VAE latents to generate a blurry base image, which can optionally be refined using SDXL's img2img functionality. Besides that, the Semantic-Level Pipeline generates a caption from EEG latents using a GIT model and employs it as a text prompt for SDXL. While this approach provides stronger semantic guidance, it is applied selectively due to the inherent instability of caption generation from EEG signals.

IV. Methodology to Assess Reproducibility, Generalizability, and Hyperparameter Robustness

This chapter outlines the study's methodology for reproducing the original ATM experiments, evaluating the model's generalizability on an external EEG dataset, and assessing its robustness to changes in key hyperparameter settings.

A. Reproducibility Assessment

Reproducibility is a fundamental principle of scientific research, ensuring that reported findings can be independently validated under equivalent conditions. In this study, the aim is to assess: "How reproducible are the results of the ATM-based EEG visual decoding framework by Li et al. [2] when tested on its original dataset?"

The reproducibility assessment focused on three criteria: Accessibility of the Original Code and Data, Complexity of Environment Setup, and Similarity of Results. Initially, the raw EEG data from all participants were preprocessed following the authors' original preprocessing pipeline to ensure consistency in signal conditioning. Subsequently, image features were extracted using a pre-trained vision-language model, and text features were generated by encoding descriptive class prompts and saved as .pt files. EEG embeddings were then computed by passing the EEG data through the ATM-S and ATM-E models, resulting in feature vectors for each EEG sample. Finally, the models were trained and evaluated based on their Top-1 and Top-5 accuracy in image retrieval tasks and their performance in k-way classification tasks. Additionally, the two-stage EEG-to-image generation pipeline was reproduced. The best resulting images were then extracted and visually compared to the original stimulus images and the corresponding reconstructions presented in the original study.

B. Generalizability Assessment

While reproducibility ensures that experimental results can be reliably replicated, generalizability concerns a model's ability to maintain performance when applied to data that differs from the conditions it was initially trained on. It is particularly important in EEG-based decoding, where signal characteristics can vary significantly across datasets due to differences in hardware, electrode placement, participant demographics, and experimental design. The research question guiding this assessment is: "To what extent can this model generalize to a new dataset (Alljoined1 [14]), and how does its performance vary when applied to EEG data with different properties, such as variations in channel configurations, signal-to-noise ratios, and stimulus types?"

The generalizability assessment focuses on three criteria: Accessibility and Compatibility of New Data, Complexity of Adapting the Experimental Pipeline, and Transferability of Performance Across Datasets. First, the raw EEG data was reshaped, and the preprocessing pipeline was modified to accommodate the structure of the new dataset. Subsequently, several internal functions within the original implementation were adapted to handle the dimensions and characteristics of the latest image set and EEG data, enabling the successful extraction of image and text features. Next, the EEG data was processed through the ATM-S and ATM-E encoders to generate EEG-image latent representations. Finally, the model's generalizability was evaluated using Top-1 accuracy, Top-5 accuracy, and k-way classification performance metrics. Due to extensive shape mismatches and hardcoded assumptions in the original codebase, the image generation component could not be executed reliably on the new dataset and was therefore excluded from the generalizability assessment.

C. Hyperparameter Robustness Assessment

Another important aspect of this study involved assessing the robustness of the proposed framework to variations in hyperparameter configurations. This evaluation was guided by the research question: {"How robust is the framework's performance when subjected to different hyperparameter configurations, and to what extent do small variations in parameters affect overall decoding accuracy?" Two main criteria addressed this: the complexity of adapting the training environment and the model's sensitivity to changes in key hyperparameters. First, the script was refactored to accept all relevant hyperparameters via command-line arguments. This change eliminated the need for manual code edits when conducting parameter

sweeps, making the experimentation process significantly more efficient and less error-prone. Second, hyperparameters were centralized in a configuration object and referenced consistently by all model components. A structured analysis was conducted by systematically varying key training parameters using a custom Python script.

V. Experiments and Results

This chapter summarizes the key findings from reproducing and extending the ATM model. It covers the accessibility of the original code and data, the complexity of the experimental setup, and the reproducibility of the models' performances. In addition, it evaluates the model's ability to generalise to a new dataset and its robustness to changes in training parameters.

A. Reproducibility Analysis

1) Accessibility of Data and Code: The code was obtained from the official GitHub repository[1]. The raw and preprocessed data and the corresponding image set are publicly available via OSF and OpenNeuro. Pretrained model weights are provided through Hugging Face to facilitate reproducibility. A comprehensive README.md file outlines the necessary steps for environment setup, and a requirements.txt file is included to assist with dependency installation. However, several essential files are missing from the repository, preventing further exploration of the low-level and semantic pipelines. No manual request is required for access.

2) Complexity of Experimental Setup: A comprehensive `README` file outlines the primary steps required to reproduce the dataset preparation, classification, and reconstruction processes. Additionally, the provided `requirements.txt` file allows for the seamless installation of all necessary packages, with no dependency conflicts encountered during setup. Most of the required code modifications were relatively straightforward, such as updating data and image paths or correcting minor syntax errors. However, as previously noted, the low-level and semantic-level pipelines could not be reproduced due to the absence of essential files, specifically `ViT-L-14_features_GIT_train.pt` and `ViT-L-14_features_GIT_test.pt`. Furthermore, while the `modeling_git` module exists, the class `GitForCausalLMClipEmb`, which is referenced in `GIT_caption_batch.ipynb`, is not implemented or publicly available, thus preventing execution of that part of the pipeline. Additionally, the notebook `Reconstruction_Metrics_ATM.ipynb`, which includes a cell responsible for displaying reconstructions alongside the ground truth images, consistently results in CUDA out-of-memory errors even with 48GB of GPU memory, twice as much as the GPU used in the original experiment. In one instance, the memory allocation attempt exceeded 11,000 GB of GPU memory. Resolving these issues would require extensive restructuring and optimization of the corresponding code segment.

[1]https://github.com/ncclab-sustech/EEG_Image_decode

3) Similarity of Results: To assess the reproducibility of classifier performance on the THINGS-EEG dataset, each model was trained in a subject-dependent setting and evaluated on Top-1, Top-5, and N-Way retrieval tasks. The results, summarized in Table I, show the mean classification accuracy and corresponding standard deviation across all subjects for the original implementation, the reproduced experiments, and other common architectures for comparison. For each metric and model, an additional row indicates the difference in performance (Reproduced – Original), with positive values highlighted in green and negative values in red to visualize reproducibility gaps. Differences in classification performance vary across models and retrieval metrics. The reproduced ATM-S model outperforms the original implementation in all reported metrics, with a Top-1 accuracy increase of +7.07% and a Top-5 improvement of +7.77%. In contrast, the reproduced ATM-E model shows moderate gains in most metrics, although a decrease of -1.31% is observed in Top-1 accuracy. These deviations are likely attributable to implementation-specific factors such as a different hardware setup affecting training dynamics.

The image reconstruction quality achieved in this experiment appears promising based on qualitative visual inspection. A side-by-side comparison of the ground truth images with those generated using the reproduced CLIP-only pipeline is shown in Figure 3a. The reconstructed images capture the overall scene structure and semantic content reasonably well. Notably, the visual results obtained using the reproduced CLIP-only implementation closely resemble those from the original CLIP-only pipeline, shown in Figure 3, regarding object layout and thematic consistency.

B. Generalisability Analysis

1) Accessibility of Data and Code: All relevant metadata as well as a script to download the shared1000 imageset, which is a subset of the Natural Scenes Dataset (NSD), a subset of the Common Objects in Context (COCO) image collection, are publicly available on GitHub[2]. Additionally, a file to map the NSD image ID to the Coco ID and a file that describes which images were shown to which subject are provided are also needed. The dataset can be accessed and downloaded via Hugging Face [3]. Compared to the THINGS-EEG2 dataset, Alljoined1 is significantly smaller, around one-eighth the size after preprocessing, and has one additional channel (using the 10-30 system for electrode placement). It includes only 80 categories, unlike the 1,654 training classes and 200 test classes in the THINGS image set. Additionally, Alljoined1 does not follow the structured image distribution of THINGS-EEG2, where each training class is represented by ten images and each test class by one. The participant pool is also reduced, with data collected from eight subjects (each contributing one or two sessions), compared to ten subjects with four sessions each in the THINGS-EEG2 dataset.

[2]https://github.com/Alljoined/alljoined-dataset1

[3]https://huggingface.co/Alljoined

Methods	2-Way	4-Way	10-Way	Top-1	Top-5
EEGITNet	76.69 ± 12.97	56.98 ± 16.31	36.35 ± 15.11	5.75 ± 3.62	18.14 ± 9.40
EEGConformer	76.17 ± 13.13	56.29 ± 16.70	34.72 ± 14.79	3.98 ± 2.80	17.10 ± 9.21
ShallowFBCSPNet	74.32 ± 12.14	53.97 ± 15.81	33.48 ± 14.35	6.10 ± 4.61	16.53 ± 9.94
EEGNetV4	92.81 ± 2.22	83.15 ± 4.20	67.81 ± 6.11	19.51 ± 5.19	48.99 ± 6.75
B.D.	78.42 ± 8.81	58.24 ± 12.13	37.97 ± 11.38	5.88 ± 3.49	18.61 ± 7.81
NICE	92.73 ± 2.75	83.26 ± 5.47	67.96 ± 8.31	19.32 ± 5.33	49.26 ± 9.69
MLP	83.09 ± 2.54	66.70 ± 4.16	45.43 ± 4.58	7.23 ± 1.66	25.14 ± 3.66
ATM-S (orig)	94.92 ± 1.45	87.91 ± 3.14	75.37 ± 5.77	28.64 ± 6.39	58.47 ± 8.97
ATM-S (repr)	96.45 ± 1.07	90.64 ± 2.23	79.84 ± 3.99	35.71 ± 5.03	66.24 ± 4.48
ATM-S (gener)	52.24 ± 2.04	26.73 ± 1.58	11.15 ± 1.36	1.13 ± 0.41	6.61 ± 1.08
ATM-S (diff)	+1.53 -42.68	+2.73 -61.18	+4.47 -64.22	+7.07 -27.51	+7.77 -51.86
ATM-E (orig)	92.99 ± 2.20	83.81 ± 4.46	68.87 ± 7.27	22.40 ± 6.62	50.59 ± 9.59
ATM-E (repr)	93.46 ± 1.47	85.93 ± 2.92	69.32 ± 3.47	21.09 ± 2.58	51.04 ± 4.11
ATM-E (gener)	47.3 ± 1.99	23.19 ± 1.64	9.29 ± 1.20	N/A	5.86 ± 1.50
ATM-E (diff)	+0.47 -45.69	+2.12 -60.62	+0.45 -59.58	+1.31 N/A	+0.45 -44.73

TABLE I: Retrieval performance averaged over all subjects. Original ATM models used batch size 1024, reproduced models used batch size 64 due to hardware constraints. 'diff' rows show performance changes from 'orig' to 'repr' (green, left) and from 'orig' to 'gener' (red, right).

2) Complexity of Experimental Setup: Before preprocessing, the raw EEG data had to be manually converted into a .npy file containing a dictionary with specific keys: raw_eeg_data, highpass, lowpass, nchan, ch_names, sfreq, and ch_type. Each key corresponds to a particular aspect of the EEG recording, such as the data array, filter settings, channel information, and sampling frequency. This structure was required to ensure compatibility with the preprocessing script. The original preprocessing pipeline was closely tailored to a specific dataset, with hardcoded channel orders and fixed assumptions regarding the number of repetitions per condition. Consequently, adapting the pipeline to new datasets was cumbersome and error-prone. Several key improvements were implemented in the revised version to enhance generalizability: the channel order can now be easily modified to accommodate different datasets, the code robustly handles variable numbers of repetitions, and the merging and saving logic has been made more flexible. These modifications substantially facilitate the adaptation of the pipeline to new experimental datasets.

To adapt the retrieval and classification pipeline for the Alljoined1 dataset, a few substantial modifications were implemented in the data loading, feature extraction, and evaluation components. The original pipeline relied on fixed assumptions regarding dataset structure, uniform repetition counts, and direct index-based alignment, which were incompatible with the less structured organization of Alljoined1. The revised data loader dynamically accommodates the dataset's new parameters, including the number of EEG channels (64), sequence length (334), and class count (80), as well as the updated directory and file structures. The adapted loader dynamically determines class labels for EEG trials using COCO category annotations and balances category assignments to handle the uneven image distribution over the classes. Image file locations are determined by session and trial indices to match a manually curated, subject-specific directory structure, and text prompts are automatically generated from an ordered list of categories. This loader robustly handles irregularities like missing data and variable trial repetitions, logging warnings, and skipping problematic entries. Additionally, the model architecture and evaluation routines were updated to reflect these dataset-specific modifications. The evaluation logic dynamically samples classes, preventing shape mismatches and ensuring accurate computation of retrieval metrics. These adaptations enabled the pipeline to process and analyze the Alljoined1 dataset successfully.

While the retrieval and classification components of the ATM framework could be effectively adapted to the Alljoined1 dataset, significant obstacles arose when attempting to adjust the generation pipeline. Successful EEG-based generative modeling, especially in supervised training like the ATM pipeline, depends on consistent and unambiguous mappings between EEG trials and their corresponding target images or image feature vectors. Within Alljoined1, however, EEG-to-image mappings were non-unique: individual images belong to multiple categories defined by COCO labels. In addition, fundamental mismatches in data dimensionality significantly impeded successful integration. Specifically, the precomputed visual feature tensors (`ViT-H-14_features_alljoined_test/training).pt`) were wrongly computed with dimensions relative to model expectations, resulting in persistent shape mismatches within the `diffusion_prior` component (DiffusionUNet). Differences in EEG channel configurations and temporal resolutions between the original and Alljoined1 datasets further complicated efforts to maintain consistent input-output dimensions, thus causing runtime errors and training instability. Consequently, despite extensive efforts to address these challenges through adjustments, the generative pipeline remained non-operational.

3) Similiarity of Results: Applying the ATM-S and ATM-E models to the Alljoined1 dataset resulted in a measurable decline in retrieval and classification performance compared to their performance on the THINGS-EEG2 dataset. On THINGS-EEG2, both models achieved consistently high Top-1, Top-5, and N-way retrieval accuracy across subjects, as shown in Table I. In contrast, when applied to Alljoined1, the same models exhibited significantly lower accuracy across all evaluated metrics. For ATM-S, Top-1 accuracy dropped from 28.64% on THINGS-EEG2 to 1.13% on Alljoined1, while

Top-5 accuracy declined from 58.47% to 6.61%. A similar trend was observed for ATM-E, with Top-1 accuracy decreasing from 21.09% to a level where no consistent measurement could be computed, and Top-5 accuracy dropping from 50.59% to 5.86%.

C. Hyperparameter Robustness Analysis

1) Complexity of Experimental Setup: Assessing the robustness of the ATM-S and ATM-E models to various hyperparameter configurations required a few modifications. Initially, both scripts were structured with many hyperparameters and architectural choices hardcoded within the source code or embedded in the configuration class. This design limited flexibility and made systematic exploration of the parameter space impractical and error-prone. To address these limitations, the script was refactored to accept all relevant hyperparameters as command-line arguments. Specifically, robustness was tested via two independent parameter sweeps: (1) jointly varying the learning rate and dropout rate across the grid [1e-4, 3e-4, 1e-3] × [0.1, 0.25, 0.5], and (2) jointly varying batch size and number of training epochs across the grid [32, 64, 128] × [20, 40, 80]. Each hyperparameter pair was examined independently to isolate its influence on the model's performance, with Top-1 accuracy and retrieval metrics systematically logged for each configuration. A custom script was written to automatically run experiments across multiple learning rates, batch sizes, and epochs to check how robust the ATM-S model is to different hyperparameter settings. This script utilized command-line arguments, making it easy to change these settings without editing the code directly. Attempts to extend the sensitivity analysis to architectural hyperparameters, such as d_model, n_heads, and e_layers, proved largely unsuccessful due to strong interdependencies within the original implementation. These efforts resulted in runtime errors and tensor shape mismatches, consequently limiting the scope of the robustness analysis to training-related hyperparameters.

2) Similiarity of Results: Hyperparameter sensitivity results reveal how classification accuracy varies across different training configurations. The outcomes of these experiments are summarized in Tables Table II and Table III, and visual representations can be found in Figures Figure 1a, Figure 2a, Figure 1b, and Figure 2b. The default values used in the original code are highlighted in grey.

Table III shows the Top-1 accuracy obtained by adjusting learning rates and dropout rates. ATM-S achieved its best accuracy (40.5%) with a learning rate of 0.0001 combined with a dropout of 0.25. Similarly, ATM-E peaked at 40.0% accuracy using a learning rate of 0.0001 but a slightly lower dropout of 0.1. Deviations from these optimal settings resulted in noticeable accuracy reductions, particularly with changes in dropout rate. The heatmaps in Figure 1a and Figure 2a visually emphasize how quickly performance can decline when hyperparameters shift away from their ideal values.

As seen in Table II, ATM-S performed best with an accuracy of 39.5% with a smaller batch size of 32 over a relatively short training period of 20 epochs. ATM-E reached optimal performance (40.0%) at a batch size of 64, also within 20 epochs. Training beyond 40 epochs generally led to either stagnant or decreasing accuracy, pointing to possible overfitting or limited further improvement. Figures Figure 1b and Figure 2b visually illustrate these patterns.

Model	Batch Size	Ep 20	Ep 40	Ep 80
ATM-E	32	36.0	31.5	30.5
	64	40.0	35.0	30.0
	128	39.5	32.0	36.5
ATM-S	32	39.5	38.0	37.0
	64	38.0	37.0	38.5
	128	36.5	39.0	37.0

TABLE II: Top-1 Accuracy (%) by Batch Size and Epochs. Original setting highlighted in grey.

Model	Learning Rate	Do 0.1	Do 0.25	Do 0.5
ATM-E	0.0001	40.0	34.5	38.0
	0.0003	35.0	29.5	33.5
	0.001	34.5	32.5	34.0
ATM-S	0.0001	38.0	40.5	35.0
	0.0003	34.5	35.5	34.0
	0.001	35.5	27.0	24.0

TABLE III: Top-1 Accuracy (%) by Learning Rate and Dropout. Original setting highlighted in grey.

VI. Discussion

The outcomes reveal the ATM framework to be highly reproducible under similar experimental settings. Both ATM-S and ATM-E models attained classification accuracies close to the initially reported outcome, even as they used differing hardware resources and made minor code modifications through addressing environment and path discrepancies. Visual outputs from the recreated CLIP-only reconstruction pipeline also emerged highly similar to the originals, reflecting consistency in learned representations. Reproducibility was nonetheless limited by the realities of working memory resources in the form of missing files for the semantic and low-level pipelines and extreme memory allocation challenges in reconstructing images.

Unlike the strong reproducibility seen on the THINGS-EEG2 dataset, both ATM-S and ATM-E suffered a dramatic reduction in performance when used on the Alljoined1 dataset. The fundamental differences between the two image sets are key contributors to this performance difference. THINGS-EEG2 contains mostly simple and clearly delineated single-object images. Alljoined1 contains complex naturalistic images often having multiple objects and cluttered backgrounds. This added visual complexity most likely introduces higher variability in the EEG signals and makes it harder for the model to identify stable and discriminating patterns. Secondly, the labeling approach employed in each dataset profoundly affects the model's performance. THINGS-EEG2 has a precise one-to-one image-class mapping, having ten images per class

in the case of training and a single image per class in the case of testing. Alljoined1 has multiple images per class, with multiple COCO categories associated, most probably diluting the correspondence between the EEG features and their visual stimuli counterparts. That can be overcome by choosing only a single category from those, but this solution doesn't capture the images' full complexity and doesn't reproduce the authors' intent. Uneven image distribution across classes further undermines performance by biasing feature extraction. To counteract this, a custom routine uses a lookup table to track how often each category is selected and swaps in less frequent classes when one appears too frequently. Other differences, like possible variations in the hardware setups or even the conditions of the recordings, may add additional incompatibilities. Since both datasets passed through the same preprocessing pipeline, that parameter can be safely discarded. Though the smaller batch size used in the replicated runs did not significantly detract from performance (as you can see from Table II, it actually improved performance), the same setup might not be sufficient when the training set used the Alljoined1 data. A smaller batch size here might have limited the capacity of the model to capture the wider signal distribution and decrease generalization capacity. These results corroborate recent studies by Yang et al. [15], highlighting the challenge in cross-domain generalization of EEG decoding models. The literature increasingly advocates training on broader datasets, applying domain adaptation methods, and developing flexible model forms that can handle neural and visual variability.

Beyond generalization challenges, findings also highlight the sensitivity of the ATM framework to hyperparameter settings. Systematic hyperparameter sweep indicated stable performance of the model in some optimum ranges—for instance, low learning rate and intermediate dropout values always resulted in superior decoding accuracy. However, performance dropped significantly when moving away from these settings, showing that although the model is not extremely sensitive, it does demand precise tuning. The effort at sweeping through architectural parameters did not succeed because of tight coupling in the initial codebase that resulted in shape mismatches and runtime errors, imposing a restriction on analyzing at the level of training level hyperparameters.

VII. Conclusion

This study tested the reproducibility, generalizability, and hyperparameter sensitivity of the Adaptive Thinking Mapper (ATM) framework. Reproduction tests on the original THINGS-EEG2 dataset included retraining classification models and image reconstruction through the given CLIP-only pipeline and comparing performance with the original implementation. Generalizability was tested by applying the trained models to the Alljoined1 dataset, which posed novel structural and semantic challenges. The last test involved a hyperparameter sensitivity analysis through variations in learning rate, dropout, batch size, and training time. The results indicated that although the essential framework proved reproducible under controlled conditions, performance decreased substantially upon transfer to more challenging datasets and became very sensitive to decoding accuracy given careful tuning. Furthermore, a lack of tunability in the included code deterred a wider exploration of architectural sensitivity. The findings highlight a gap between strong in-domain performance and real-application feasibility and indicate the necessity for future efforts to focus on modular model design, wider dataset benchmarking, and thorough cross-domain testing.

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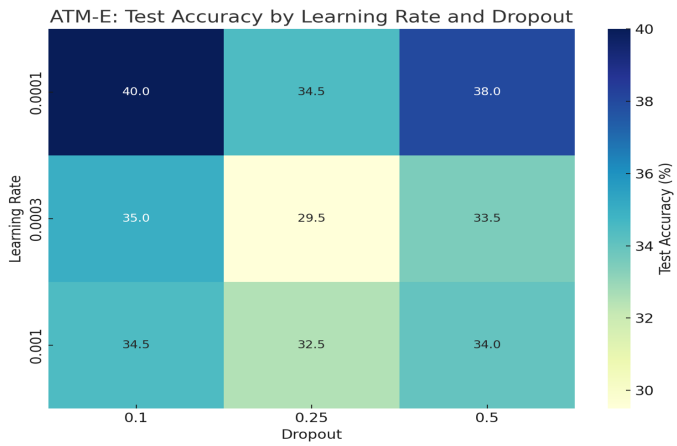
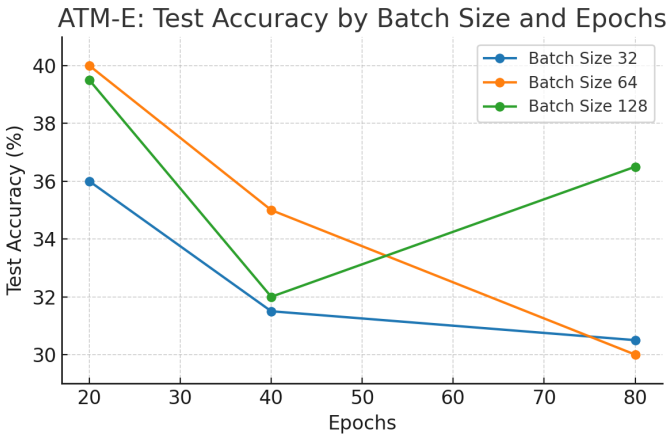
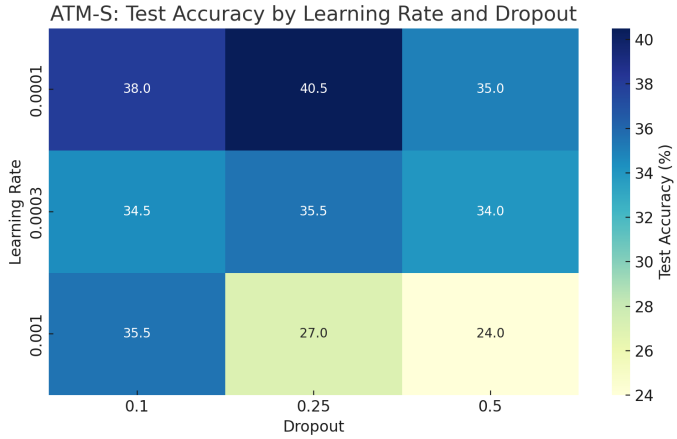


(a) Top-1 accuracy for different learning rates and dropout values (ATM-E).

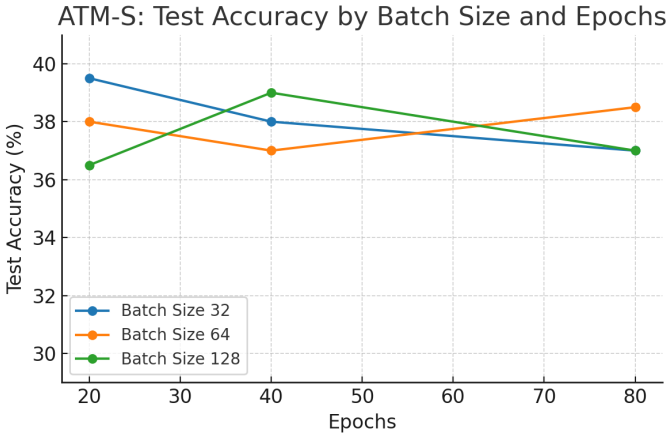


(b) Top-1 accuracy across different batch sizes and epochs (ATM-E).

Fig. 1: Hyperparameter robustness analysis for ATM-E on subject 08.

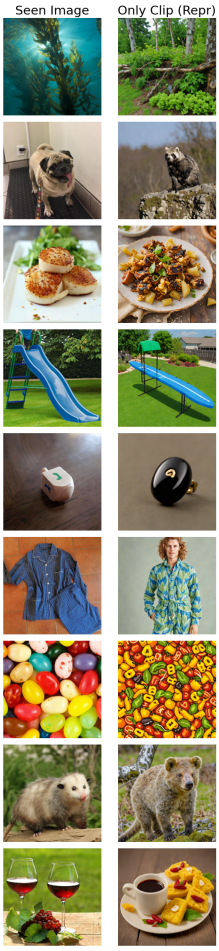


(a) Top-1 accuracy for different learning rates and dropout values (ATM-S).

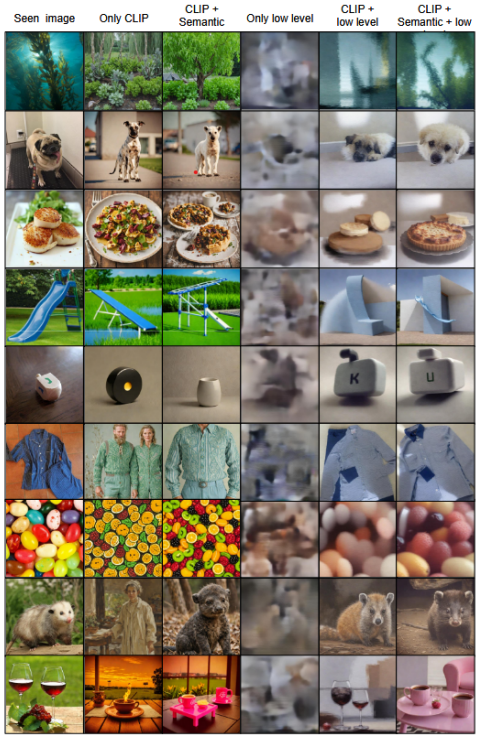


(b) Top-1 accuracy across different batch sizes and epochs (ATM-S).

Fig. 2: Hyperparameter robustness analysis for ATM-S on subject 08.



(a) Side-by-side comparison of the original seen images (left column) and their reconstructions using the reproduced CLIP-only pipeline (right column).



(b) Reconstruction comparison across six model variants. From left to right: seen image, CLIP-only, CLIP+semantic, low-level only, CLIP+low-level, and CLIP+semantic+low-level. The full model achieves the most coherent and perceptually accurate reconstructions. Taken from [2].

Fig. 3: Visual comparison of EEG-based image reconstructions for Subject 8. (a) Reproduced CLIP-only pipeline. (b) Original reconstructions from different model configurations.